

indexing description:

$w_{ij}^{[n]}$ $\rightarrow$ weight connecting j -th feature of layer $n-1$ to the i -th node of layer n

$z_k^{[n]}(i)$ $\rightarrow$ z value of i -th training example in unit number k of layer # n
(linear value)

$a_k^{[n]}(i)$ $\rightarrow$ activation value of i -th training example in unit number k of layer # n
(node)

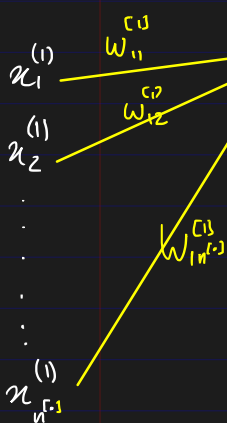
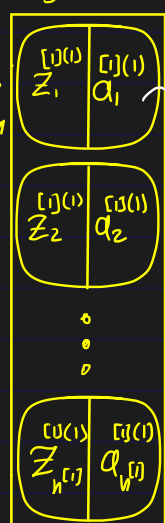
$x_k^{(i)}$ $\rightarrow$ k -th feature of i -th training example

$y^{(i)}$ $\rightarrow$ predicted value for i -th training example

Forward Propagation:

layer - dims = $[n^{[0]}, n^{[1]}, n^{[2]}, \dots, n^{[L]}]$

layer #1

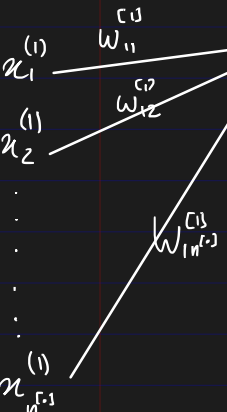
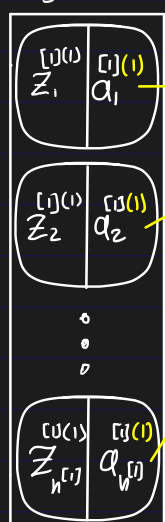


$$z_1^{[1]} = [w_{11}^{[1]} \ w_{12}^{[1]} \ \dots \ w_{1n^{[0]}}^{[1]}] \begin{bmatrix} x_1^{(1)} \\ x_2^{(1)} \\ \vdots \\ x_{n^{[0]}}^{(1)} \end{bmatrix} + b_1^{[1]} = w_{[0,:]}^{[1]} x^{(1)} + b_1^{[1]}$$

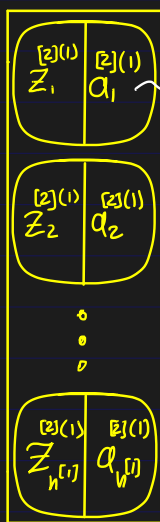
$$a_1^{[1]} = g(z_1^{[1]})$$

$$W^{[1]} = \begin{bmatrix} w_{11}^{[1]} & w_{12}^{[1]} & \dots & w_{1n^{[0]}}^{[1]} \\ w_{21}^{[1]} & w_{22}^{[1]} & \dots & w_{2n^{[0]}}^{[1]} \\ \vdots & \vdots & \ddots & \vdots \\ w_{n^{[1]}1}^{[1]} & w_{n^{[1]}2}^{[1]} & \dots & w_{n^{[1]}n^{[0]}}^{[1]} \end{bmatrix}_{n^{[1]} \times n^{[0]}}$$

layer #1



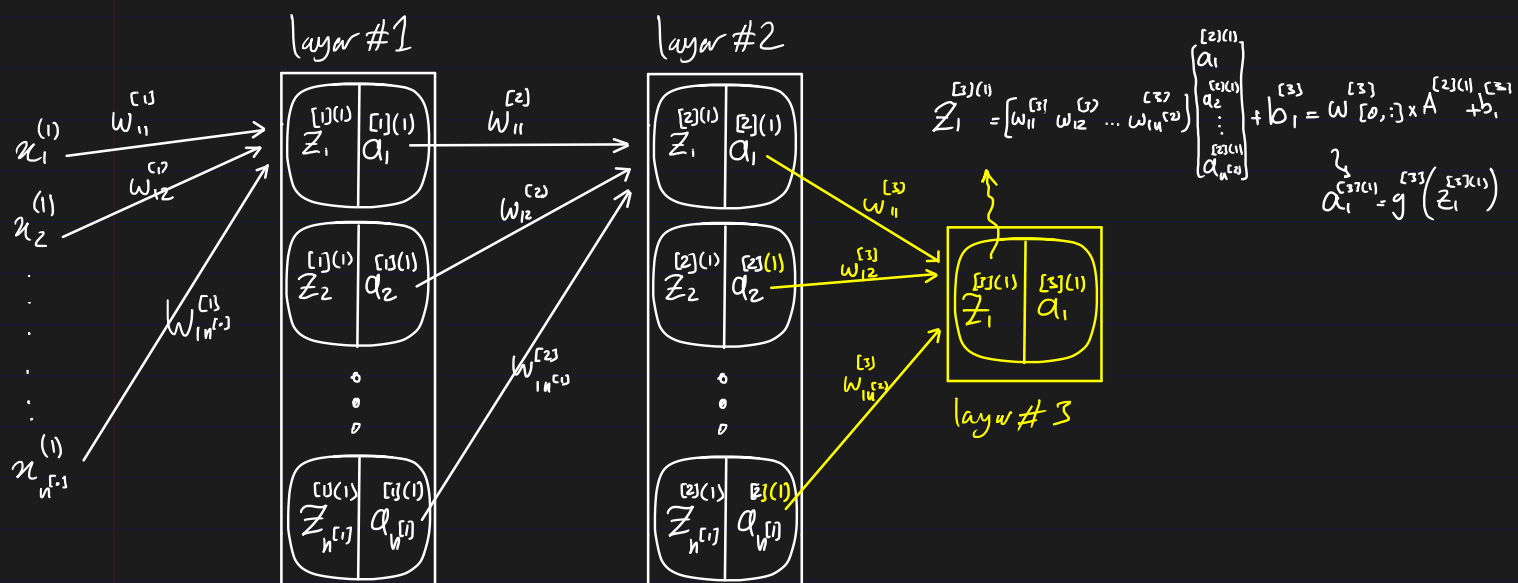
layer #2



$$z_1^{[2]} = [w_{11}^{[2]} \ w_{12}^{[2]} \ \dots \ w_{1n^{[1]}}^{[2]}] \begin{bmatrix} a_1^{[1]} \\ a_2^{[1]} \\ \vdots \\ a_{n^{[1]}}^{[1]} \end{bmatrix} + b_1^{[2]}$$

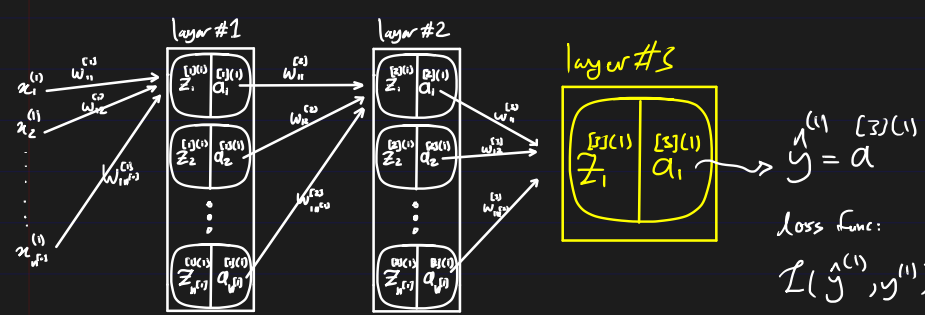
$$= w_{[0,:]}^{[2]} A^{[1]} + b_1^{[2]}$$

$$a_1^{[2]} = g(z_1^{[2]})$$



$$\begin{bmatrix} z_1^{(2)} \\ z_2^{(2)} \\ \vdots \\ z_n^{(2)} \end{bmatrix} = \begin{bmatrix} w_{11}^{(2)} & w_{12}^{(2)} & \dots & w_{1n}^{(2)} \end{bmatrix} \begin{bmatrix} a_1^{(1)} \\ a_2^{(1)} \\ \vdots \\ a_n^{(1)} \end{bmatrix} + b_1^{(2)}$$

$$a_i^{(l)} = g(z_i^{(l)})$$



$$\hat{y}^{(1)} = a_1^{(3)}$$

Loss function:

$$\mathcal{L}(\hat{y}^{(1)}, y^{(1)}) = - (y^{(1)} \times \log \hat{y}^{(1)} + (1 - y^{(1)}) \times \log (1 - \hat{y}^{(1)}))$$

-Vectorization:

$$W^{[l]} = \begin{bmatrix} w_{11}^{[l]} & w_{12}^{[l]} & \dots & w_{1n}^{[l-1]} \\ w_{21}^{[l]} & w_{22}^{[l]} & \dots & w_{2n}^{[l-1]} \\ \vdots & \vdots & \ddots & \vdots \\ w_{n1}^{[l]} & w_{n2}^{[l]} & \dots & w_{nn}^{[l-1]} \end{bmatrix}_{n \times n}$$

$$X = \begin{bmatrix} | & | & | & | \\ x^{(1)} & x^{(2)} & \dots & x^{(m)} \\ | & | & | & | \end{bmatrix}_{n \times m}$$

training examples

$$Z^{[1]} = W^{[1]} X + b^{[1]} \rightarrow Z^{[1]}_{n \times m} = \begin{bmatrix} z_1^{(1)} & z_1^{(2)} & \dots & z_1^{(m)} \\ z_2^{(1)} & z_2^{(2)} & \dots & z_2^{(m)} \\ \vdots & \vdots & \ddots & \vdots \\ z_n^{(1)} & z_n^{(2)} & \dots & z_n^{(m)} \end{bmatrix}$$

$$A^{[1]} = g(Z^{[1]}) \rightsquigarrow A^{[1]}_{n \times m}$$

more generally:

$$\text{if we set } A^{(0)}_{n \times m} = X_{n \times m}$$

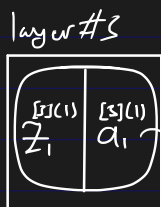
$$Z^{[l]} = W^{[l]} \times A^{[l-1]} + b^{[l]}$$

$$A^{[l]} = g(Z^{[l]})$$



Backpropagation and Gradients:

Starting from
Output layer:



$$\hat{y} = a$$

Loss func:

$$\mathcal{L}(\hat{y}^{(1)}, y^{(1)}) = -(y^{(1)} \times \log \hat{y}^{(1)} + (1 - y^{(1)}) \times \log(1 - \hat{y}^{(1)}))$$

$$\frac{d\mathcal{L}}{da_1^{[3](1)}} \checkmark$$

$$\frac{d\mathcal{L}}{dz_1^{[3](1)}} \checkmark$$

$$\frac{d\mathcal{L}}{da_1^{[3](1)}} = \frac{-y^{(1)}}{a_1^{[3](1)}} + \frac{1-y^{(1)}}{1-a_1^{[3](1)}} \rightarrow \text{we call this } \delta a_1^{[3](1)}$$

$$\frac{d\mathcal{L}}{dz_1^{[3](1)}} = \frac{d\mathcal{L}}{da_1^{[3](1)}} \times \frac{da_1^{[3](1)}}{dz_1^{[3](1)}} = \delta a_1^{[3](1)} \times \frac{d}{dz_1^{[3](1)}} (g^{[3]}(z_1^{[3](1)})) \rightarrow \text{we call this } \delta z_1^{[3](1)}$$

finding $\frac{d\mathcal{L}}{dw_{11}^{[3]}}$, $\frac{d\mathcal{L}}{db_1^{[3]}}$:

$$\frac{d\mathcal{L}}{dw_{11}^{[3]}} = \frac{d\mathcal{L}}{dz_1^{[3](1)}} \times \frac{dz_1^{[3](1)}}{dw_{11}^{[3]}}$$

$$= \delta z_1^{[3](1)} \times \frac{d}{dw_{11}^{[3]}} (w_{11}^{[3]} \times a_1^{[2](1)} + w_{12}^{[3]} \times a_2^{[2](1)} + \dots + w_{1n}^{[3]} \times a_n^{[2](1)} + b_1^{[3]})$$

$$= \delta z_1^{[3](1)} \times a_1^{[2](1)}$$

Also:

$$\frac{d\mathcal{L}}{dw_{12}^{[3]}} = \delta z_1^{[3](1)} \times a_2^{[2](1)}$$

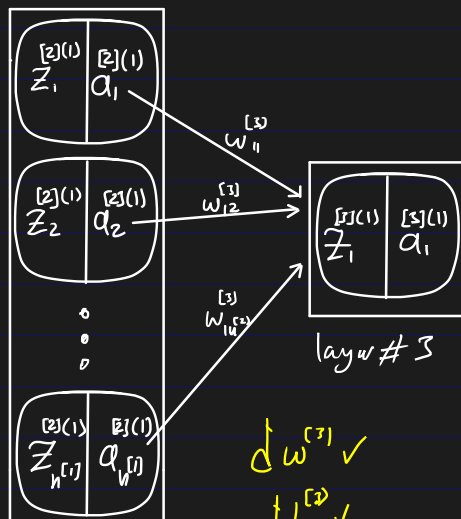
} move
generally:

$$\frac{d\mathcal{L}}{dw_{ij}^{[3]}} = \delta z_1^{[3](1)} \times a_j^{[2](1)}$$

we call this
 $\delta w_{ij}^{[3]}$

$$\frac{d\mathcal{L}}{db_1^{[3]}} = \frac{d\mathcal{L}}{dz_1^{[3](1)}} \times \frac{dz_1^{[3](1)}}{db_1^{[3]}} = \delta z_1^{[3](1)}$$

layer #2



$$\delta w_{ij}^{[3]} \checkmark$$

$$\delta b_1^{[3]} \checkmark$$

finding $\delta a_1^{[2](1)}$:

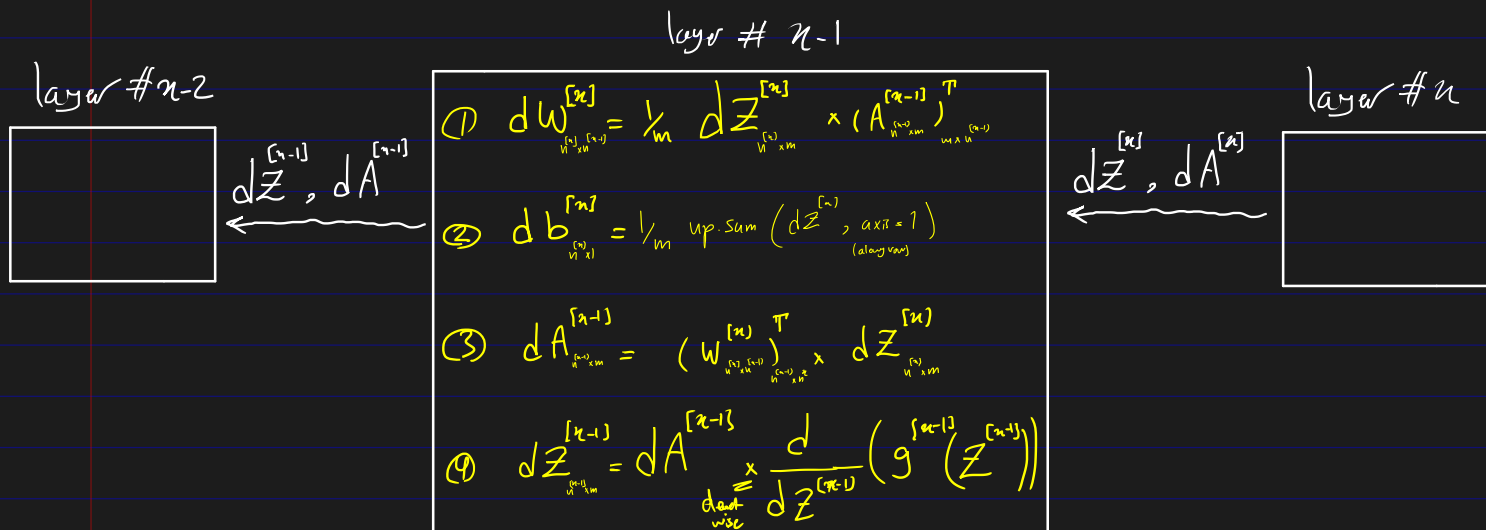
$$\frac{d\mathcal{L}}{da_1^{[2](1)}} = \frac{d\mathcal{L}}{dz_1^{[3](1)}} \times \frac{dz_1^{[3](1)}}{da_1^{[2](1)}} = \delta z_1^{[3](1)} \times \frac{d}{da_1^{[2](1)}} (w_{11}^{[3]} \times a_1^{[2](1)} + \dots + w_{1n}^{[3]} \times a_n^{[2](1)} + b_1^{[3]}) = \delta z_1^{[3](1)} \times w_{11}^{[3]} \rightarrow \delta a_1^{[2](1)}$$

move generally: $\delta a_j^{[2](1)} = \delta z_1^{[3](1)} \times w_{1j}^{[3]} \rightarrow \delta a_j^{[2](1)}$

Vectorizing:

in the last layer: $dA^{[L]} = \frac{-y}{A^{[L]}} + \frac{1-y}{1-A^{[L]}}$, $dZ^{[L]} = dA^{[L]} \times \frac{d}{dZ^{[L]}}(g^{[L]}(Z^{[L]}))$
($l=L$)

in hidden layers:



Reasoning for ①:

- we calculated that $dW_{ij}^{[n]} = dZ_i^{[n]} \times a_j^{[n-1]}$

i -th input from layer $[n-1]$ $\rightarrow dW_{ij}^{[n]} = dZ_i^{[n]} \times a_j^{[n-1]}$

(vectorizing) for all training exmples $\rightarrow A^{[n-1]} = \begin{bmatrix} \end{bmatrix}_{n^{[n-1]} \times m}$, $dZ^{[n]} = \begin{bmatrix} \end{bmatrix}_{n^{[n]} \times m}$, $dW^{[n]} = \begin{bmatrix} \end{bmatrix}_{n^{[n]} \times n^{[n-1]}}$

Reasoning for ②: ~~8/16~~ ET

Reasoning for ③:

we know: $dq_j^{[n-1]} = dZ_i^{[n]} \times dW_{ij}^{[n]}$

$dA_{i,1:m}^{[n-1]}$, $dZ_{i,1:m}^{[n]}$, $dW_{i,1:n}^{[n]}$